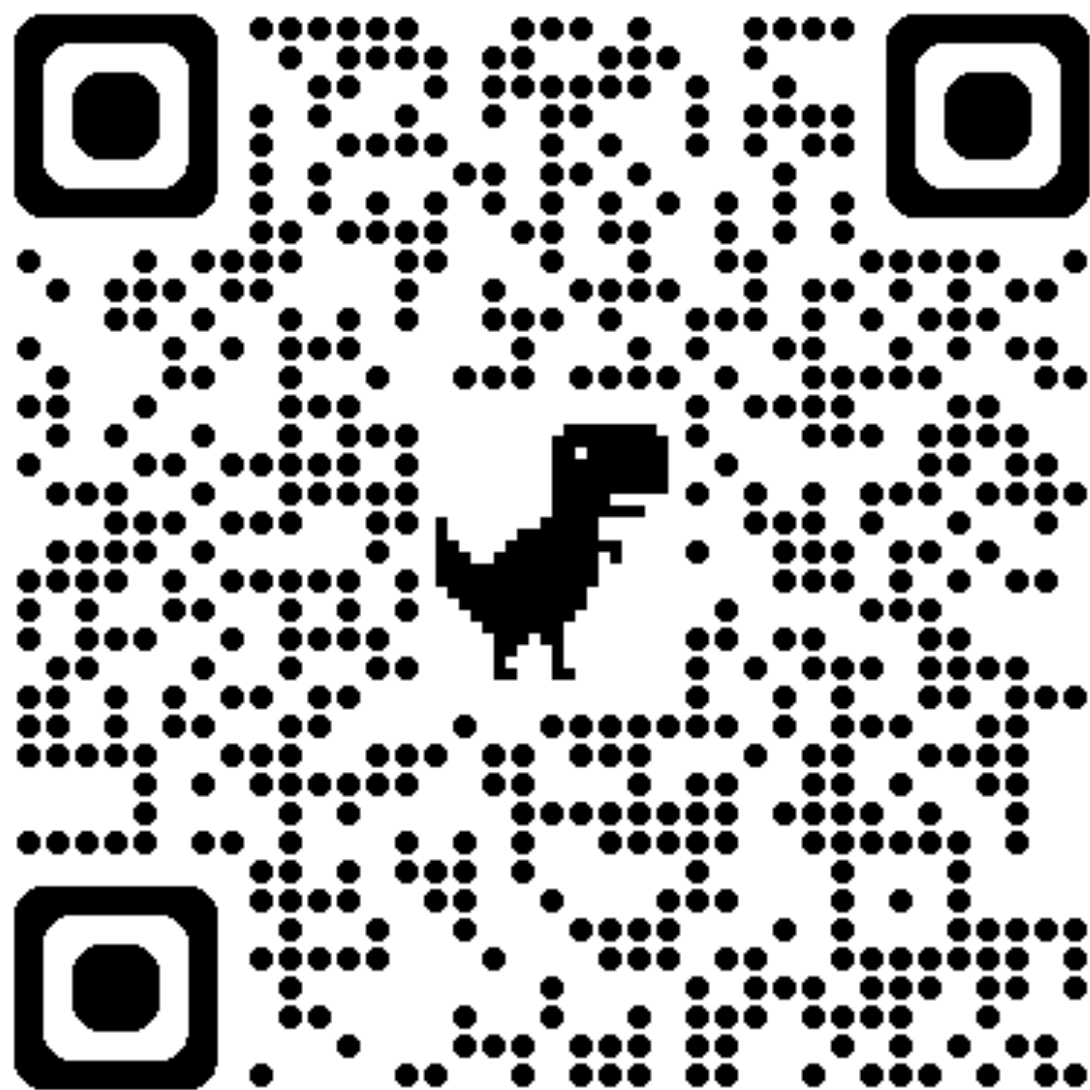
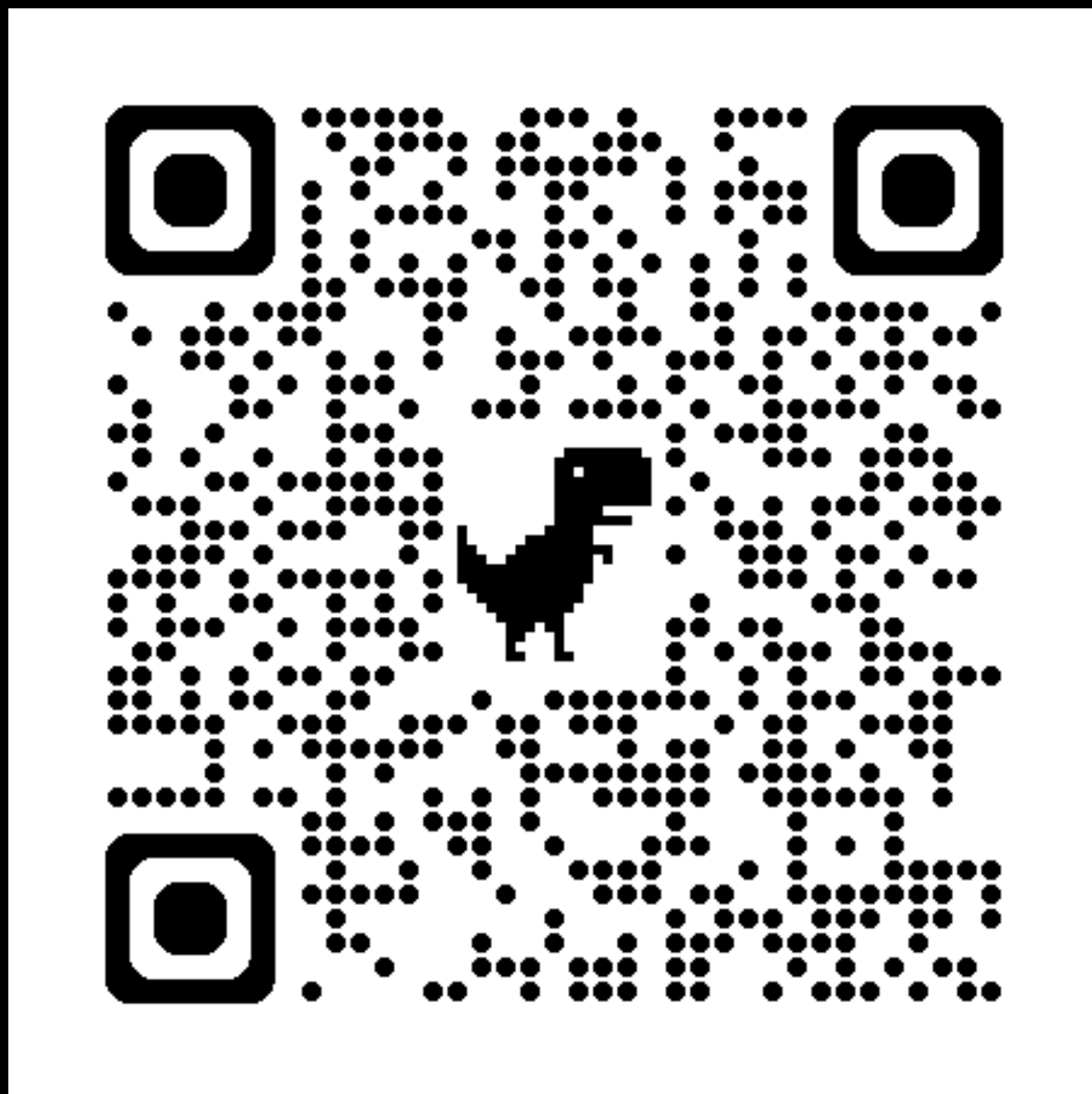




Interactive Demo: Kantorovich

Implement code to:

1. Turn $\mathbf{a}$, $\mathbf{b}$, $\mathbf{C}$ into LP objective and constraints for an LP solver
2. Recover π from LP solution and compare to Monge solution



Standard form:

$$\begin{aligned} \min_{\pi \in \mathbb{R}^{n \times m}} \quad & \sum_{i=1}^n \sum_{j=1}^m C_{ij} \pi_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^m \pi_{ij} = a_i \quad \forall i = 1, \dots, n, \\ & \sum_{i=1}^n \pi_{ij} = b_j \quad \forall j = 1, \dots, m, \\ & \pi_{ij} \geq 0 \quad \forall i, j. \end{aligned}$$

Vectorized form:

$$\begin{aligned} \min_{x \in \mathbb{R}^{nm}} \quad & c^\top x \\ \text{s.t.} \quad & (I_n \otimes \mathbf{1}_m^\top) x = a, \\ & (\mathbf{1}_n^\top \otimes I_m) x = b, \\ & x \geq 0. \end{aligned}$$

Interactive Demo: Kantorovich